

## Module 5

### Controllable, Reachable, and Invariant Sets

1

- In this module, we will address the following basic questions.
- For a given system:
  - *From what sets of states can I reach a given target set?*
  - *What states can I get to from a specified initial set?*
  - *From what set of states can I start my system and guarantee that constraints will never be violated?*

2

## Some Math Definitions

3

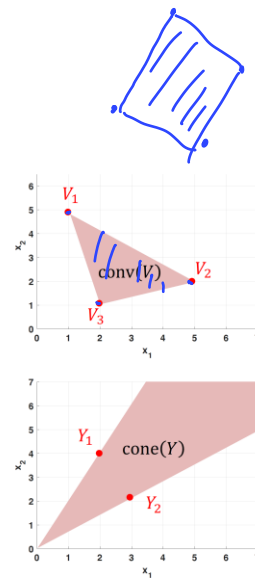
### Convex Hulls & Cones

**Definition:** The convex hull of a finite set of points  $V \subset \mathbb{R}^n$  is the set of all convex combinations of points in  $V$  and is denoted by  $\text{conv}(V)$ :

$$\text{conv}(V) = \left\{ x \in \mathbb{R}^n : x = \sum_{i=1}^k \lambda_i V_i, 0 \leq \lambda_i \leq 1, \sum_{i=1}^k \lambda_i = 1 \right\}, \text{ where } V = \{V_1, V_2, \dots, V_k\}$$

**Definition:** A cone spanned by a finite set of points  $Y \subset \mathbb{R}^n$  is defined as

$$\text{cone}(Y) = \left\{ \sum_{i=1}^{k'} \lambda_i Y_i, \lambda_i \geq 0, i = 1, 2, \dots, k' \right\}, \text{ where } Y = \{Y_1, Y_2, \dots, Y_{k'}\}$$



4

# Minkowski Sum

**Definition:** The **Minkowski Sum** of two sets  $\mathcal{P}, \mathcal{Q} \subset \mathbb{R}^n$  is defined as

$$\mathcal{P} \oplus \mathcal{Q} = \{x + y : x \in \mathcal{P}, y \in \mathcal{Q}\}.$$

Geometric intuition:

- Slide one set around every point of the other set
- Take the union of all those shifted copies

$$(i) \quad A = \{(1, 2)\} \quad B = \{(3, 4)\} \quad A \oplus B = \{(4, 6)\}$$

$$(ii) \quad A = \{(x, y) \in \mathbb{R}^2 \mid 1 \leq x \leq 2, 0 \leq y \leq 2\}$$

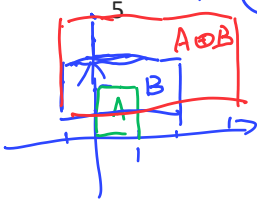
$$A \oplus B = \{(x, y) \in \mathbb{R}^2 \mid 3 \leq x \leq 4, 1 \leq y \leq 3\} \leftarrow \text{TRANSLATION OF SET}$$



$$(iii) \quad A = [1, 3] \quad B = [4, 5] \quad A \oplus B = [5, 8]$$

$$(iv) \quad A = [0, 1] \times [0, 2] \quad B = [-1, 2] \times [1, 3]$$

$$A \oplus B = ([0, 1] \oplus [-1, 2]) \times ([0, 2] \oplus [1, 3]) \\ = [-1, 3] \times [1, 5]$$



## Polyhedra

$$\mathcal{X} = \{x \in \mathbb{R}^n \mid A x \leq b\} \quad \mathcal{U} = \{u \in \mathbb{R}^n \mid A_u u \leq b_u\}$$

A Polyhedron is defined by two equivalent representations.

$$m = \# \text{ constraints} \\ n = \#(x)$$

**Definition:** A **polyhedron**  $\mathcal{P}$  in  $\mathbb{R}^n$  is the intersection of a *finite* set of closed halfspaces in  $\mathbb{R}^n$ :

$$\mathcal{P} = \{x \in \mathbb{R}^n : Ax \leq b\}, \quad A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m.$$

**Definition:** A **V-polyhedron**  $\mathcal{P}$  in  $\mathbb{R}^n$  is the Minkowski sum of the convex hull of a finite set of points  $V = \{V_1, V_2, \dots, V_k\}$  in  $\mathbb{R}^n$ , and the cone generated by a finite set of points  $Y = \{Y_1, Y_2, \dots, Y_{k'}\}$  in  $\mathbb{R}^n$ :

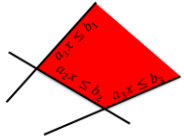
$$\mathcal{P} = \text{conv}(V) \oplus \text{cone}(Y).$$



# Polytopes

**Definition:** A **polytope** is a bounded polyhedra

A  $\mathcal{H}$ -polytope is a bounded  $\mathcal{H}$ -polyhedron, i.e., it does not contain any ray  $\{x + ty : t \geq 0\}$ .

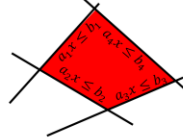


**An unbounded  $\mathcal{H}$ -polyhedron:**

$$\mathcal{P} = \{x \in \mathbb{R}^n : Ax \leq b\}$$

$$A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix},$$

$a_i$  are rows of  $A$ ,  
 $b_i$  are components of  $b$



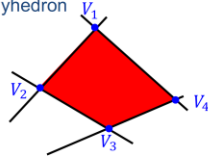
**A  $\mathcal{H}$ -polytope:**

$$\mathcal{P} = \{x \in \mathbb{R}^n : Ax \leq b\}$$

$$A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}, b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix},$$

$a_i$  are rows of  $A$ ,  
 $b_i$  are components of  $b$

A  $\mathcal{V}$ -polytope is a bounded  $\mathcal{V}$ -polyhedron



**A  $\mathcal{V}$ -polytope:**

$$\mathcal{P} = \text{conv}(V),$$

$$V = \{V_1, V_2, V_3, V_4\}$$

**Note:** Polytopes and polyhedral are always convex

7

## Operations on Polytopes and Polyhedra

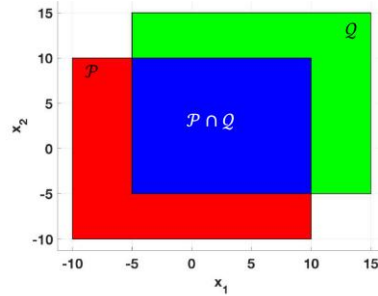
8



# Intersection

**Definition:** The *intersection* of two polytopes  $\mathcal{P}$  and  $\mathcal{Q} \subset \mathbb{R}^n$  is defined as

$$\mathcal{P} \cap \mathcal{Q} = \{x : x \in \mathcal{P} \text{ and } x \in \mathcal{Q}\}$$



9

$$\begin{bmatrix} x \\ y \end{bmatrix} \quad \begin{matrix} x \in \mathbb{R}^n \\ y \in \mathbb{R}^m \end{matrix}$$

## Projection

**Definition:** Given a polytope  $\mathcal{P}$ :

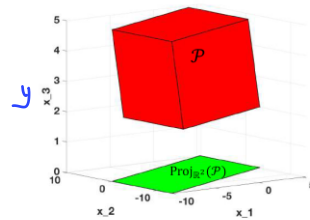
$$\mathcal{P} = \{[x^T y^T]^T \in \mathbb{R}^{n+m} : \underbrace{P^x x + P^y y}_{A} \leq \underbrace{P^c}_{b}\} \subset \mathbb{R}^{n+m},$$

The **projection onto the  $x$ -space  $\mathbb{R}^n$**  is defined as

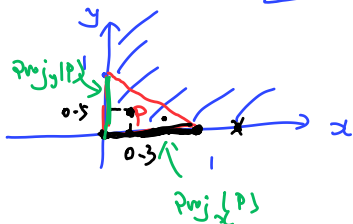
$$\text{proj}_x(\mathcal{P}) = \{x \in \mathbb{R}^n : \exists y \in \mathbb{R}^m \text{ s.t. } P^x x + P^y y \leq P^c\}.$$

↑  
THAT EXISTS

$$n=1 \quad m=1$$



$$(i) \quad \mathcal{P} = \{[x \ y]^T \mid x \geq 0, y \geq 0, x+y \leq 1\}$$



$$\begin{aligned} \text{Proj}_x(\mathcal{P}) &= \{x \in \mathbb{R} \mid \exists y, x \geq 0, y \geq 0, x+y \leq 1\} \\ &= \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}. \end{aligned}$$

$$(ii) \quad x(k+1) = x(k) + u(k), \quad \text{where } x^+ = x + u$$

10

$$(14) \leq 1 \quad |x^+| \leq 2$$

FEASIBLE SET: THE SET OF ALL CURRENT STATE  
 $x(k)$  OR  $x$  OF WHICH THERE

2. FINDS AN ADMISSIBLE INPUT  $u(k)$  OR  $u$ , SUCH THAT THE NEXT STATE  $x^+$  OR  $x(k+1)$  SATISFIES THE STATE CONSTRAINT.

$$|u| \leq 1$$

$$|x^+| \leq 2$$

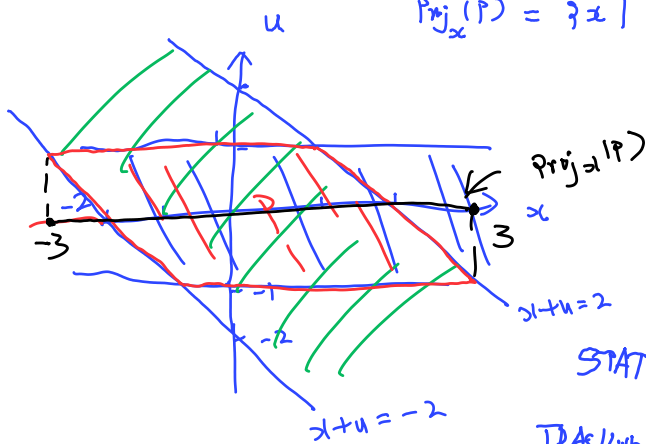
$$P = \{x, u \mid -1 \leq u \leq 1, -2 \leq x^+ \leq 2\}$$

$$x^+ = x + u$$

$$\text{Proj}_x(P) = \{x \mid \exists u, -1 \leq u \leq 1, -2 \leq x + u \leq 2\}$$

$$= \{x \mid -3 \leq x \leq 3\}$$

ROBUST CONSTRAINT TIGHTENING.



STATE CONSTRAINTS  $x = \{x \in \mathbb{R}^2 \mid |x_1| \leq 5, |x_2| \leq 5\}$

TRACKING ERROR  $E = \{e \in \mathbb{R}^2 \mid |e_1| \leq 1, |e_2| \leq 0.5\}$

TRUE STATE  $x = z + e$  : WE WANT  $x \in X$  FOR ALL  $e \in E$ .

11

TIGHTENED ROBUST CONSTRAINTS

$$z \in X \ominus E = \{z \mid \forall e \in E, z + e \in X\}$$

## Set Differences

**Definition:** The **set difference** of two sets  $P, Q \subset \mathbb{R}^n$  is defined as

$$P \setminus Q = \{x \in \mathbb{R}^n \mid x \in P, x \notin Q\}.$$

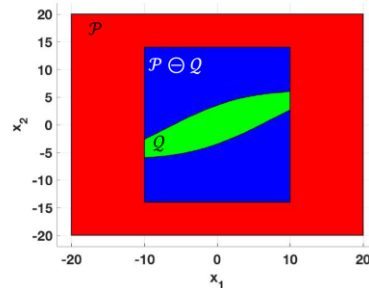
**Definition:** The **Pontryagin difference** of two sets  $P, Q \subset \mathbb{R}^n$  is defined as

$$P \ominus Q = \{x \in \mathbb{R}^n \mid x + q \in P, \forall q \in Q\}.$$

**Note:** The Pontryagin difference defined above is also sometimes called the *P-difference* or the *Minkowski difference*.

**Geometric intuition:**

- Slide the set  $Q$  around inside  $P$
- The  $P$ -difference is the set of all locations of the reference point where the whole shifted copy still fits inside  $P$



12

13

$$x^+ = Ax + Bu$$

$$x(k+1) = Ax(k) + Bu(k)$$

## Preimage of a Polyhedron Under an Affine Map

Let  $\mathcal{P}$  be a polyhedron and  $f(z)$  be an affine mapping:

$$\mathcal{P} = \{x \in \mathbb{R}^n : P^x x \leq P^c\}, \quad P^x \in \mathbb{R}^{n_p \times n}, \quad P^c \in \mathbb{R}^{n_p}$$

$$f: z \in \mathbb{R}^m \rightarrow Az + b, \quad A \in \mathbb{R}^{m_A \times m}, \quad b \in \mathbb{R}^{m_A}$$

**Definition:** Let  $m_A = n$ . The **composition of  $\mathcal{P}$  and  $f$**  is defined as the following polyhedron:

$$\mathcal{P} \circ f = \{z \in \mathbb{R}^m : P^x f(z) \leq P^c\} = \{z \in \mathbb{R}^m : P^x Az \leq P^c - P^x b\}.$$

$$\mathcal{X} = \{(x_1, x_2) \mid \begin{matrix} x_1 \leq 1 \\ x_2 \leq 2 \end{matrix}\} \leftarrow \text{CONSTRAINTS ON STATE}$$

$$x^+ = f = Ax + b, \quad A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Ax + b \in \mathcal{X}$$

$$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \left( \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \leq \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow \mathcal{P} \circ f = \{x \in \mathbb{R}^2 \mid$$

$$\begin{matrix} x_1 + 2x_2 \leq 0 \\ -x_1 + x_2 \leq 2 \end{matrix}\}$$

14

15

## Image of a Polyhedron Under an Affine Map

**Definition:** Let  $m = n$ . The **composition of  $f$  and  $\mathcal{P}$**  is defined as the following polyhedron:

$$f \circ \mathcal{P} = \{y \in \mathbb{R}^m : \exists x \in \mathcal{P} \text{ and } y = Ax + b\}.$$

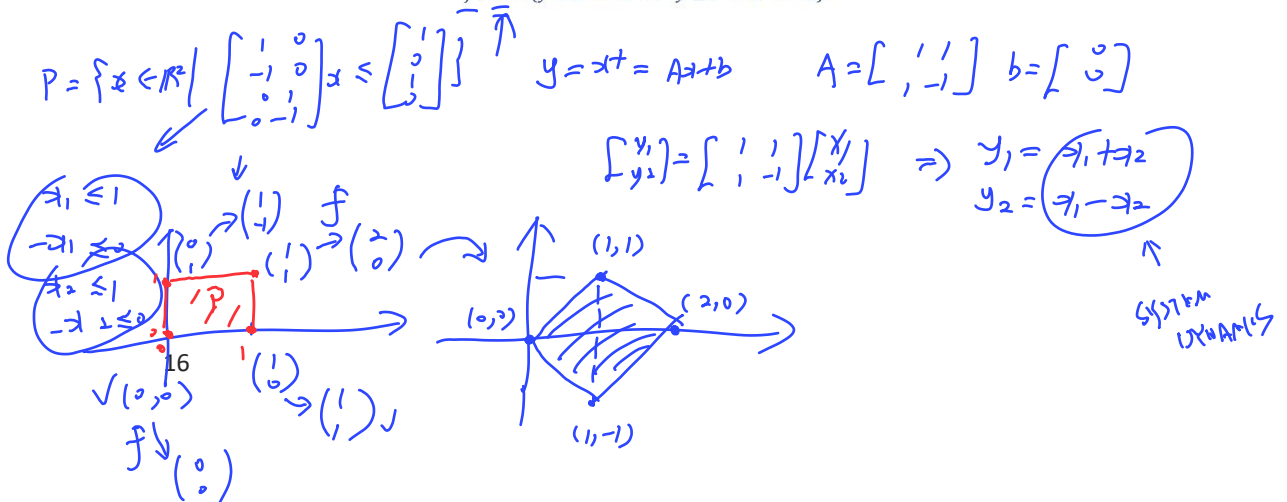
**Calculation:** The polyhedron  $f \circ \mathcal{P}$  can be calculated as follows:

**General Case:** Express  $\mathcal{P}$  as  $\mathcal{P} = \text{conv}(V)$ ,  $V = \{V_1, V_2, \dots, V_k\}$ , then

$$f \circ \mathcal{P} = \text{conv}(F), \text{ where } F = \{AV_1 + b, AV_2 + b, \dots, AV_k + b\}.$$

**Special Case:** if  $m_A = m = n$  and  $A$  is invertible,  $x = A^{-1}y - A^{-1}b$ , and it follows that

$$f \circ \mathcal{P} = \{y \in \mathbb{R}^m : P^x A^{-1}y \leq P^c + P^x A^{-1}b\}.$$



17

## Matlab commands using MPT3 toolbox

### To define a polyhedral object:

Using the  $\mathcal{H}$  representation: `P=Polyhedron('A',A,'B',b);`  
 Using the  $\mathcal{V}$  representation: `P=Polyhedron('V',V);` ( $V$  is a matrix with rows corresponding to vertices of  $P$ )  
 To plot: `plot(P)`

### To extract information about a given polyhedron:

Extract  $\mathcal{H}$  representation: `A=P.A, b=P.b`  
 Extract the  $\mathcal{V}$  representation: `V=P.V`

### To calculate Minkowski sum and Pontryagin Difference for polyhedral $P$ and $Q$ :

$P \oplus Q$ : `P+Q`  
 $P \ominus Q$ : `P-Q`

### To calculate the intersection of $P$ and $Q$ :

$P \cap Q$ : `P.intersect(Q)`

### To calculate the projection of polyhedron $P \in \mathbb{R}^n$ onto $\mathbb{R}^2$ :

$\text{Proj}_{\mathbb{R}^2}(P)$ : `projection(P,[1 2])`

**Multi-Parametric Toolbox 3:** <https://www.mpt3.org/>

- Please check out the quadcopter and power network examples there!
- Can be extended to the MPC setting!
- You should consider using it if you are dealing with linear or piecewise affine systems.

18

## Controllable and Reachable Sets

19

### System definitions

Consider two types of discrete-time systems:

Autonomous systems:

$$x(t+1) = f_a(x(t)), \quad (1)$$

Systems subject to external inputs:

$$x(t+1) = f(x(t), u(t)). \quad (2)$$

Subject to state and input constraints:

$$x(t) \in \mathcal{X}, \quad u(t) \in \mathcal{U}, \quad \text{for all } t \geq 0 \quad (3)$$

$\mathcal{X}$  and  $\mathcal{U}$  are polyhedral sets.

**Note:** For a state-feedback controller  $u = g(x)$ , a system (2) can be written in autonomous form (1).

$$\begin{aligned} u(t) &= g(x(t)) \\ x(t+1) &= f(x(t), g(x(t))) = f_a(x(t)) \end{aligned}$$

20

$$x(k+1) = Ax(k)$$

## Precursor Set Definition

**Definition:** For the autonomous system (1), the **precursor set** to a set  $\mathcal{S}$  is defined as

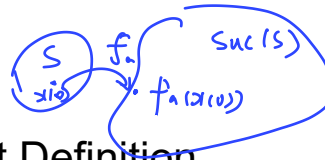
$$\text{Pre}(\mathcal{S}) = \{x \in \mathbb{R}^n : f_a(x) \in \mathcal{S}\} = \underline{S \circ A} = S \circ f$$

**Definition:** For the system (2) with inputs, the **precursor set** to a set  $\mathcal{S}$  is defined as

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ \text{Pre}(\mathcal{S}) &= \{x \in \mathbb{R}^n : \exists u \in \mathcal{U} \text{ such that } f(x, u) \in \mathcal{S}\} \\ &= (S \oplus (-B) \circ u) \circ A \end{aligned}$$

The set  $\text{Pre}(\mathcal{S})$  is the set of all states that evolve into, or can be driven into, the target set  $\mathcal{S}$  in one time step.

21



## Successor Set Definition

**Definition:** For the autonomous system (1), the **successor set** from a set  $\mathcal{S}$  is defined as

$$\text{Suc}(\mathcal{S}) = \{x \in \mathbb{R}^n : \exists x(0) \in \mathcal{S} \text{ such that } x = f_a(x(0))\} = f \circ S = A \circ S$$

**Definition:** For the system (2) with inputs, the **successor set** from a set  $\mathcal{S}$  is defined as

$$\begin{aligned} \text{Suc}(\mathcal{S}) &= \{x \in \mathbb{R}^n : \exists x(0) \in \mathcal{S}, \exists u(0) \in \mathcal{U} \text{ such that } x = f(x(0), u(0))\} \\ &= (A \circ S) \oplus (B \circ u) \end{aligned}$$

All states contained in  $\mathcal{S}$  are mapped into the set  $\text{Suc}(\mathcal{S})$  under the map  $f_a$ , or under the map  $f$  for some input,  $u \in \mathcal{U}$ .

22

## N-step Controllable Set Definition

↓  
BACKWARD

**Definition:** For a given target set  $S \subseteq \mathcal{X}$ , the **N-step controllable set**  $\mathcal{K}_N(S)$  of the system (1) or (2) subject to constraints (3) is defined recursively as

$x \in \mathcal{X}, u \in \mathcal{U}$

$$\mathcal{K}_j(S) = \text{Pre}(\mathcal{K}_{j-1}(S)) \cap \mathcal{X}, \quad \mathcal{K}_0(S) = S, \quad j = 1, 2, \dots, N$$

All states of the system (1) belonging to  $\mathcal{K}_N(S)$  will evolve to the target set  $S$  in  $N$  steps while satisfying state constraints.

All states of the system (2) belonging to  $\mathcal{K}_N(S)$  can be driven to the target set  $S$  in  $N$  steps, with a suitable control sequence, while satisfying input and state constraints.

$$\begin{aligned} k_0(S) &= S \\ k_1(S) &= \text{Pre}(k_0(S)) \cap \mathcal{X} = \text{Pre}(S) \cap \mathcal{X} \\ k_2(S) &= \text{Pre}(k_1(S)) \cap \mathcal{X} \end{aligned}$$

23

## N-step Reachable Set Definition

↓  
FORWARD

**Definition:** For a given initial set  $X_0 \subseteq \mathcal{X}$ , the **N-step Reachable set**  $\mathcal{R}_N(X_0)$  of the system (1) or (2) subject to constraints (3) is defined as

$$\mathcal{R}_{i+1}(X_0) = \text{Suc}(\mathcal{R}_i(X_0)) \cap \mathcal{X}, \quad \mathcal{R}_0(X_0) = X_0, \quad i = 0, 1, \dots, N-1$$

The system (1) can reach any state in  $\mathcal{R}_N(X_0)$  in  $N$ -steps, starting from some state in  $X_0$ , while satisfying state constraints.

The system (2) can be driven to any state in  $\mathcal{R}_N(X_0)$  in  $N$ -steps, starting from some state in  $X_0$ , while satisfying state and input constraints.

$$\begin{aligned} \mathcal{R}_0(X_0) &= X_0 \\ \mathcal{R}_1(X_0) &= \text{Suc}(X_0) \cap \mathcal{X} \\ \mathcal{R}_2(X_0) &= \text{Suc}(\mathcal{R}_1(X_0)) \cap \mathcal{X} \end{aligned}$$

24

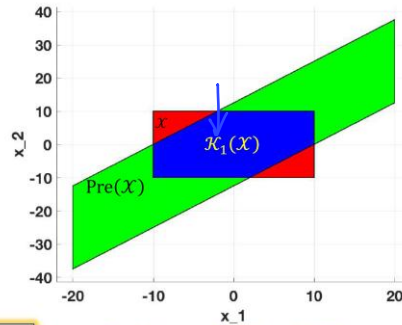




The **1-step controllable set to the set  $\mathcal{X}$**  is calculated as follows:

$$\mathcal{K}_1(\mathcal{X}) = \text{Pre}(\mathcal{X}) \cap \mathcal{X}$$

$$\mathcal{K}_1(\mathcal{X}) = \left\{ x : \begin{bmatrix} -0.5 & 0 \\ -1 & 0.8 \\ 0.5 & 0 \\ 1 & -0.8 \\ 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} x \leq \begin{bmatrix} 10 \\ 10 \\ 10 \\ 10 \\ 10 \\ 10 \\ 10 \\ 10 \end{bmatrix} \right\}$$



The 1-step controllable set  $\mathcal{K}_1(\mathcal{X})$

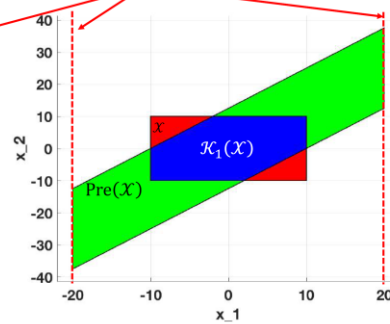
*All states of the system belonging to  $\mathcal{K}_1(\mathcal{X})$  will evolve to the target set  $\mathcal{X}$  in 1 step while satisfying state constraints.*

27

The following representation contains redundant constraints:

$$\mathcal{K}_1(\mathcal{X}) = \left\{ x : \begin{bmatrix} -0.5 & 0 \\ -1 & 0.8 \\ 0.5 & 0 \\ 1 & -0.8 \\ 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} x \leq \begin{bmatrix} 10 \\ 10 \\ 10 \\ 10 \\ 10 \\ 10 \\ 10 \\ 10 \end{bmatrix} \right\}$$

**Redundant Constraints**



The **minimal representation** for  $\mathcal{K}_1(\mathcal{X})$  is given by

$$\mathcal{K}_1(\mathcal{X}) = \left\{ x : \begin{bmatrix} -1 & 0.8 \\ 1 & -0.8 \\ 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} x \leq \begin{bmatrix} 10 \\ 10 \\ 10 \\ 10 \\ 10 \\ 10 \end{bmatrix} \right\}$$

**Note:** for this example,  $\mathcal{K}_N(\mathcal{X}) = \mathcal{K}_1(\mathcal{X})$  for  $N \geq 1$

28

The **successor set from the set  $\mathcal{X}$**  is calculated as follows:

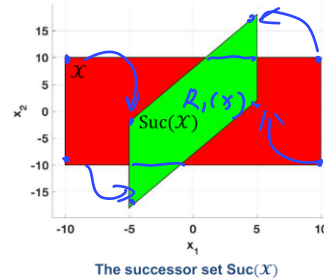
Express  $\mathcal{X}$  as a  $\mathcal{V}$ -polyhedron:

$$\mathcal{X} = \text{conv}(\mathcal{V}), \quad \mathcal{V} = \{V_1, V_2, V_3, V_4\} = \left\{ \begin{bmatrix} 10 \\ 10 \end{bmatrix}, \begin{bmatrix} 10 \\ -10 \end{bmatrix}, \begin{bmatrix} -10 \\ 10 \end{bmatrix}, \begin{bmatrix} -10 \\ -10 \end{bmatrix} \right\}$$

Apply the map  $A$  to each vertex of  $\mathcal{X}$ :

$$\text{Suc}(\mathcal{X}) = A \circ \mathcal{X} = \text{conv}(AV_1, AV_2, AV_3, AV_4)$$

$$\text{Suc}(\mathcal{X}) = \text{conv} \left( \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 18 \end{bmatrix}, \begin{bmatrix} -5 \\ -18 \end{bmatrix}, \begin{bmatrix} -5 \\ -2 \end{bmatrix} \right)$$



**All states contained in  $\mathcal{X}$  are mapped into the set  $\text{Suc}(\mathcal{X})$  under the map  $A$**

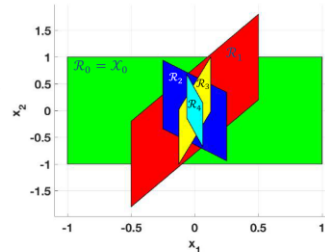
29

Consider a set  $\mathcal{X}_0 = \left\{ \begin{bmatrix} -1 \\ -1 \end{bmatrix} \leq x \leq \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} = \text{conv} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix} \right\}$ .

The **N-step reachable sets from the set  $\mathcal{X}_0$**  are calculated using:

$$\mathcal{R}_{i+1}(\mathcal{X}_0) = \text{Suc}(\mathcal{R}_i(\mathcal{X}_0)) \cap \mathcal{X}, \quad \mathcal{R}_0(\mathcal{X}_0) = \mathcal{X}_0, \quad i = 0, 1, \dots, N-1$$

$$\begin{aligned} \mathcal{R}_0(\mathcal{X}_0) &= \mathcal{X}_0 \quad \downarrow \\ \mathcal{R}_1(\mathcal{X}_0) &= \text{Suc}(\mathcal{X}_0) \cap \mathcal{X} \\ \mathcal{R}_2(\mathcal{X}_0) &= \text{Suc}(\mathcal{R}_1(\mathcal{X}_0)) \cap \mathcal{X} \\ \mathcal{R}_3(\mathcal{X}_0) &= \text{Suc}(\mathcal{R}_2(\mathcal{X}_0)) \cap \mathcal{X} \\ \mathcal{R}_4(\mathcal{X}_0) &= \text{Suc}(\mathcal{R}_3(\mathcal{X}_0)) \cap \mathcal{X} \end{aligned}$$



Each successor set can be calculated using the method shown on the previous slide.

**The system can reach any state in  $\mathcal{R}_N(\mathcal{X}_0)$  in  $N$ -steps, starting from some state in  $\mathcal{X}_0$ , while satisfying state constraints.**

30

## Example: System with Inputs

Consider the following second order unstable system

$$x(t+1) = f(x(t), u(t)) = Ax(t) + Bu(t) = \begin{bmatrix} 1.25 & 0 \\ 1 & -1.5 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t),$$

Subject to state and input constraints

$$x(t) \in \mathcal{X} = \left\{ \begin{bmatrix} -10 \\ -10 \end{bmatrix} \leq x \leq \begin{bmatrix} 10 \\ 10 \end{bmatrix} \right\},$$

$$u(t) \in \mathcal{U} = \{-5 \leq u \leq 5\}$$

The constraint sets can be written as  $\mathcal{H}$ -polytopes:

$$\mathcal{X} = \{x : Hx \leq h\}, \quad \mathcal{U} = \{u : H_u u \leq h_u\}$$

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad h = \begin{bmatrix} 10 \\ 10 \\ 10 \\ 10 \end{bmatrix}, \quad H_u = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad h_u = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

31

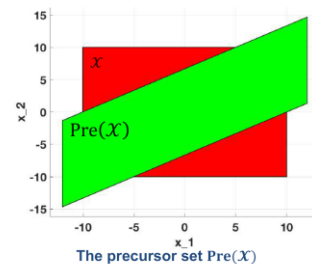
The **precursor set for the set  $\mathcal{X}$**  is calculated as follows:

$$\begin{aligned} \text{Pre}(\mathcal{X}) &= \{x \in \mathbb{R}^2 : \exists u \in \mathcal{U} \text{ s.t. } Ax + Bu \in \mathcal{X}\} \\ &= \left\{ x \in \mathbb{R}^2 : \exists u \in \mathbb{R} \text{ s.t. } \begin{bmatrix} HA & HB \\ 0 & H_u \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} \leq \begin{bmatrix} h \\ h_u \end{bmatrix} \right\} \end{aligned}$$

**Note:** The  $\text{Pre}(\mathcal{X})$  is the projection of the polytope defined by  $\begin{bmatrix} HA & HB \\ 0 & H_u \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} \leq \begin{bmatrix} h \\ h_u \end{bmatrix}$  onto the state space (in this example,  $\mathbb{R}^2$ ).

$\text{Pre}(\mathcal{X})$ , written in  $\mathcal{H}$ -polyhedron form is:

$$\begin{bmatrix} -5/6 & 0 \\ 5/6 & 0 \\ 1 & -1.5 \\ -1 & 1.5 \end{bmatrix} x \leq \begin{bmatrix} 10 \\ 10 \\ 10 \\ 10 \end{bmatrix}$$



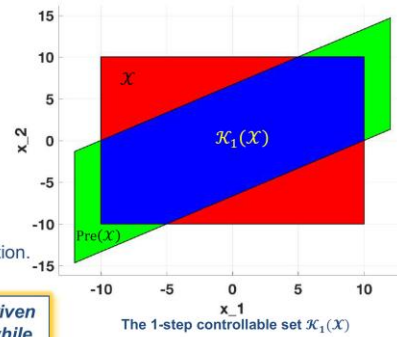
32

The **1-step controllable set to the set  $\mathcal{X}$**  is calculated as follows:

$$\mathcal{K}_1(\mathcal{X}) = \left\{ x : \begin{bmatrix} -1 & 1.5 \\ 1 & -1.5 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} x \leq \begin{bmatrix} 10 \\ 10 \\ 10 \\ 10 \\ 10 \end{bmatrix} \right\}$$

**Note:** The representation above is the minimal representation.

*All states of the system belonging to  $\mathcal{K}_1(\mathcal{X})$  can be driven to the target set  $\mathcal{X}$  in 1 step, with a suitable control, while satisfying input and state constraints.*



33

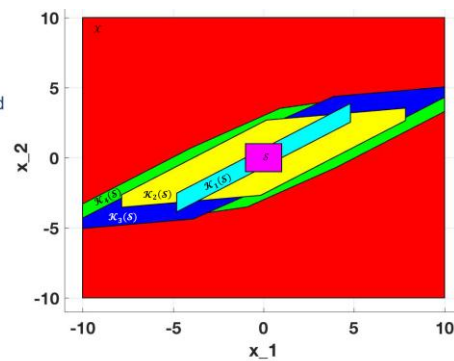
Consider a set  $\mathcal{S} = \left\{ \begin{bmatrix} -1 \\ -1 \end{bmatrix} \leq x \leq \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

The **N-step controllable sets to the set  $\mathcal{S}$**  are calculated as follows:

$$\mathcal{K}_j(\mathcal{S}) = \text{Pre}(\mathcal{K}_{j-1}(\mathcal{S})) \cap \mathcal{X}, \quad \mathcal{K}_0(\mathcal{S}) = \mathcal{S}, \\ j = 1, 2, \dots, N$$

$$\begin{aligned} \mathcal{K}_0(\mathcal{S}) &= \mathcal{S} \\ \mathcal{K}_1(\mathcal{S}) &= \text{Pre}(\mathcal{S}) \cap \mathcal{X} \\ \mathcal{K}_2(\mathcal{S}) &= \text{Pre}(\mathcal{K}_1(\mathcal{S})) \cap \mathcal{X} \\ \mathcal{K}_3(\mathcal{S}) &= \text{Pre}(\mathcal{K}_2(\mathcal{S})) \cap \mathcal{X} \\ \mathcal{K}_4(\mathcal{S}) &= \text{Pre}(\mathcal{K}_3(\mathcal{S})) \cap \mathcal{X} \end{aligned}$$

*All states of the system belonging to  $\mathcal{K}_N(\mathcal{S})$  can be driven to the target set  $\mathcal{S}$  in  $N$  steps, with a suitable control sequence, while satisfying input and state constraints.*



34

The **successor set from the set  $\mathcal{X}$**  is calculated as follows:

Express  $\mathcal{X}$  as a  $\mathcal{V}$ -polyhedron:

$$\mathcal{X} = \text{conv}(V), \quad V = \{V_1, V_2, V_3, V_4\} = \left\{ \begin{bmatrix} 10 \\ 10 \end{bmatrix}, \begin{bmatrix} 10 \\ -10 \end{bmatrix}, \begin{bmatrix} -10 \\ 10 \end{bmatrix}, \begin{bmatrix} -10 \\ -10 \end{bmatrix} \right\}$$

Apply the map  $A$  to each vertex of  $\mathcal{X}$ :

$$A \circ \mathcal{X} = \text{conv}(AV_1, AV_2, AV_3, AV_4)$$

Now, consider the effect of the input  $u \in \mathcal{U}$ :

$$\text{Suc}(\mathcal{X}) = \{y + Bu : y \in A \circ \mathcal{X}, u \in \mathcal{U}\}$$

Note that the set  $\{z : z = Bu, u \in \mathcal{U}\}$  is the composition  $B \circ \mathcal{U}$ ,

$$\mathcal{U} = \text{conv}(V_u), \quad V_u = \{V_{u1}, V_{u2}\} = \{-5, 5\}, \quad B \circ \mathcal{U} = \text{conv}(BV_{u1}, BV_{u2})$$

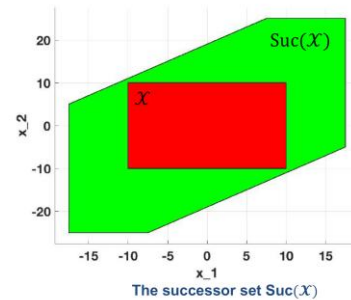
35

$$\text{Suc}(\mathcal{X}) = \{y + Bu : y \in A \circ \mathcal{X}, u \in \mathcal{U}\}$$

Note that the set  $\{z : z = Bu, u \in \mathcal{U}\}$  is the composition  $B \circ \mathcal{U}$ .

$$\text{Suc}(\mathcal{X}) = \{y + z : y \in A \circ \mathcal{X}, z \in B \circ \mathcal{U}\}$$

$$\text{Suc}(\mathcal{X}) = (A \circ \mathcal{X}) \oplus (B \circ \mathcal{U})$$



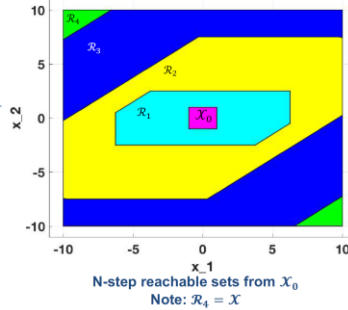
36

Consider a set  $X_0 = \left\{ \begin{bmatrix} -1 \\ -1 \end{bmatrix} \leq x \leq \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} = \text{conv} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix} \right\}$ .

The **N-step reachable sets** from the set  $X_0$  are calculated using:

$$\mathcal{R}_{i+1}(X_0) = \text{Suc}(\mathcal{R}_i(X_0)) \cap \mathcal{X}, \quad \mathcal{R}_0(X_0) = X_0, \quad i = 0, 1, \dots, N-1$$

$$\begin{aligned} \mathcal{R}_0(X_0) &= X_0 \\ \mathcal{R}_1(X_0) &= \text{Suc}(X_0) \cap \mathcal{X} \\ \mathcal{R}_2(X_0) &= \text{Suc}(\mathcal{R}_1(X_0)) \cap \mathcal{X} \\ \mathcal{R}_3(X_0) &= \text{Suc}(\mathcal{R}_2(X_0)) \cap \mathcal{X} \\ \mathcal{R}_4(X_0) &= \text{Suc}(\mathcal{R}_3(X_0)) \cap \mathcal{X} \end{aligned}$$



Each successor set can be calculated using the method shown on the previous slides.

*The system can be driven to any state in  $\mathcal{R}_N(X_0)$  in  $N$ -steps, starting from some state in  $X_0$ , while satisfying state and input constraints.*

37

## Summary: Predecessor & Successor Sets for Linear Systems Without Disturbances (using MPT3)

**Linear Autonomous Systems**  
 $x(t+1) = Ax(t), x(t) \in \mathcal{X}$  for all  $t \geq 0$ ,  
 $\mathcal{X} = \{x : Hx \leq h\}$

|                      |                       |
|----------------------|-----------------------|
| <b>Set</b>           | <b>Operation</b>      |
| Pre( $\mathcal{X}$ ) | $\mathcal{X} \circ A$ |
| Suc( $\mathcal{X}$ ) | $A \circ \mathcal{X}$ |

**MATLAB**

```
model=LTISystem('A',A);
X=Polyhedron('A',H,'b',h);
Xpre=model.reachableSet('X',X,'direction','backward');
Xsuc=model.reachableSet('X',X,'direction','forward');
figure
plot(Xpre)
figure
plot(Xsuc)
```

**Linear Systems with Inputs**  
 $x(t+1) = Ax(t) + Bu(t), \quad x(t) \in \mathcal{X}, \quad u(t) \in \mathcal{U}$  for all  $t \geq 0$ ,  
 $\mathcal{X} = \{x : Hx \leq h\}, \quad \mathcal{U} = \{u : Hu \leq h_u\}$

|                      |                                                       |
|----------------------|-------------------------------------------------------|
| <b>Set</b>           | <b>Operation</b>                                      |
| Pre( $\mathcal{X}$ ) | $(\mathcal{X} \oplus (-B \circ \mathcal{U})) \circ A$ |
| Suc( $\mathcal{X}$ ) | $(A \circ \mathcal{X}) \oplus (B \circ \mathcal{U})$  |

**MATLAB**

```
model = LTISystem('A',A,'B',B);
X=Polyhedron('A',H,'b',h);
U=Polyhedron('A',Hu,'b_u',hu);
Xpre = model.reachableSet('X', X, 'U', 'direction', 'backward');
Xsuc = model.reachableSet('X', X, 'U', 'direction', 'forward');
figure
plot(Xpre)
figure
plot(Xsuc)
```

For more MATLAB examples, see the code for examples in Ch. 10 of "Predictive Control for Linear and Hybrid Systems" by Borrelli, Bemporad and Morari, available here: <http://www.mpc.berkeley.edu/mpc-course-material>

38

Autonomous system
Controlled system

↓                      ↗

## Invariant and Control Invariant Sets

39

### (Positively) Invariant sets

$$x(k+1) = f_a(x(k)) \quad x(k+1) = Ax(k)$$

Positively Invariant sets **apply to autonomous systems of the form (1).**

Positively Invariant sets are useful in solving the following problem:

For a given feedback controller,  $u = g(x)$ , find the set of initial states whose trajectory will never violate constraints.

$$x(k+1) = f(x(k), u(k))$$

$$u(k) = g(x(k))$$

$$= f(x(k), g(x(k)))$$

$$= f_a(x(k))$$

40



$$x(k+1) = Ax(k)$$

**Definition:** A set  $\mathcal{O} \subseteq \mathcal{X}$  is **positively invariant** for the autonomous system (1) subject to constraints (3) if

$$x(0) \in \mathcal{O} \Rightarrow x(t) \in \mathcal{O}, \quad \text{for all } t \in \mathbb{N}_+.$$

**Definition:** The set  $\mathcal{O}_\infty \subseteq \mathcal{X}$  is the **maximal positively invariant set** of the autonomous system (1) subject to the constraints (3) if

1.  $\mathcal{O}_\infty$  is positively invariant, and
2.  $\mathcal{O}_\infty$  contains all of the positively invariant sets contained in  $\mathcal{X}$ .

**Note:** Sometimes positively invariant sets are referred to simply as *invariant sets*.

41

**Theorem** (Geometric Condition for Invariance): A set  $\mathcal{O}$  is a positively invariant set if and only if  $\mathcal{O} \subseteq \text{Pre}(\mathcal{O})$ .

**Proof:** We must prove both *necessity* ( $\mathcal{O}$  is positively invariant  $\Rightarrow \mathcal{O} \subseteq \text{Pre}(\mathcal{O})$ ), and *sufficiency* ( $\mathcal{O} \subseteq \text{Pre}(\mathcal{O}) \Rightarrow \mathcal{O}$  is positively invariant). Both proofs are by contradiction.

*Necessity:* Suppose  $\mathcal{O}$  is positively invariant but  $\mathcal{O} \not\subseteq \text{Pre}(\mathcal{O})$ . Since  $\mathcal{O} \not\subseteq \text{Pre}(\mathcal{O})$ , there exists a  $\bar{x} \in \mathcal{O}$  such that  $\bar{x} \notin \text{Pre}(\mathcal{O})$ . From the definition of  $\text{Pre}(\mathcal{O})$ ,  $f_a(\bar{x}) \notin \mathcal{O}$ , and therefore  $\mathcal{O}$  is not positively invariant; a contradiction.  $\square$

*Sufficiency:* Suppose  $\mathcal{O} \subseteq \text{Pre}(\mathcal{O})$  but  $\mathcal{O}$  is not positively invariant. Since  $\mathcal{O}$  is not positively invariant, then there exists  $\bar{x} \in \mathcal{O}$  such that  $f_a(\bar{x}) \notin \mathcal{O}$ . This implies that  $\bar{x} \in \mathcal{O}$  and  $\bar{x} \notin \text{Pre}(\mathcal{O})$ , and therefore  $\mathcal{O} \not\subseteq \text{Pre}(\mathcal{O})$ ; a contradiction.  $\square$

**Note:** The condition  $\mathcal{O} \subseteq \text{Pre}(\mathcal{O})$  in the Theorem above is equivalent to  $\text{Pre}(\mathcal{O}) \cap \mathcal{O} = \mathcal{O}$ , i.e.,

$$\mathcal{O} \subseteq \text{Pre}(\mathcal{O}) \Leftrightarrow \text{Pre}(\mathcal{O}) \cap \mathcal{O} = \mathcal{O}$$

42

**Algorithm 1:** Computation of  $\mathcal{O}_\infty$ 

**Input:**  $f_a, \mathcal{X}$  ←  $\{x | 4x \leq bx\}$   
**Output:**  $\mathcal{O}_\infty$

1. Set  $k = 0$ .
2. Let  $\Omega_k = \mathcal{X}$ ,
3. Let  $\Omega_{k+1} = \text{Pre}(\Omega_k) \cap \Omega_k$
4. If  $\Omega_{k+1} = \Omega_k$ , then  $\mathcal{O}_\infty \leftarrow \Omega_{k+1}$
5. Else, increment  $k = k + 1$ , go to Step 3.

This algorithm generates a set sequence  $\{\Omega_k\}$  satisfying  $\Omega_{k+1} \subseteq \Omega_k$  for all  $k \in \mathbb{N}$ , and terminates when  $\Omega_{k+1} = \Omega_k$  so that  $\Omega_k$  is the maximal positively invariant set  $\mathcal{O}_\infty$  for  $x(t+1) = f_a(x(t))$  subject to constraints.

**Notes:**

- If  $\Omega_k = \emptyset$  for some  $k$ , then  $\mathcal{O}_\infty = \emptyset$ .
- In general, the algorithm may never terminate. If this is the case, it can be shown that  $\mathcal{O}_\infty = \lim_{k \rightarrow \infty} \Omega_k$ .
- If the algorithm does terminate, then the set  $\mathcal{O}_\infty$  is said to be finely determined.
  - Sufficient conditions for finite determination of  $\mathcal{O}_\infty$ :  $f_a(x)$  is linear and stable, and the constraint set  $\mathcal{X}$  is bounded and contains the origin.

**MATLAB:**

```
model = LTISystem('A',A);
X=Polyhedron('A',H,'b','h');
model.x.with('setConstraint');
model.x.setConstraint=X;
Oinf=model.invariantSet();
```

→  $k$ : determinant number.

43

## Example

Consider the following second order autonomous, stable system

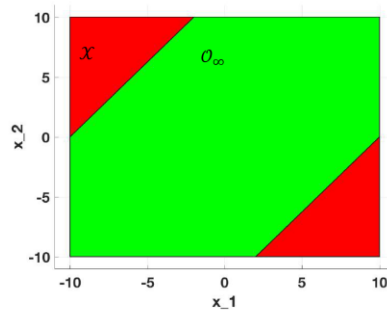
$$x(t+1) = f_a(x(t)) = Ax(t) = \begin{bmatrix} 0.5 & 0 \\ 1 & -0.8 \end{bmatrix} x(t),$$

Subject to state constraints

$$x(t) \in \mathcal{X} = \left\{ \begin{bmatrix} -10 \\ -10 \end{bmatrix} \leq x \leq \begin{bmatrix} 10 \\ 10 \end{bmatrix} \right\}.$$

The maximal positively invariant set is

$$\mathcal{O}_\infty = \left\{ x : \begin{bmatrix} 1 & -0.8 \\ -1 & 0.8 \\ 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} x \leq \begin{bmatrix} 10 \\ 10 \\ 10 \\ 10 \\ 10 \\ 10 \end{bmatrix} \right\}$$



The maximal positive invariant set

**Note:** For this example, Algorithm 1 terminates after only 1 iteration, and  $\mathcal{O}_\infty = \Omega_1 = \text{Pre}(\mathcal{X}) \cap \mathcal{X}$

44

# Control Invariant Sets

Control Invariant sets *apply to systems with external inputs of the form (2).*  $x(k+1) = f(x(k), u(k))$

These sets are useful in solving the following problem:

*Find the set of initial states for which there exists a controller such that the system constraints are never violated.*

45

**Definition:** A set  $\mathcal{C} \subseteq \mathcal{X}$  is **control invariant** for the system (2) subject to constraints (3) if

$$x(0) \in \mathcal{C} \Rightarrow \exists u(t) \in \mathcal{U} \text{ s.t. } f(x(t), u(t)) \in \mathcal{C}, \quad \text{for all } t \in \mathbb{N}_+.$$

**Definition:** The set  $\mathcal{C}_\infty \subseteq \mathcal{X}$  is the **maximal control invariant set** of the system (2) subject to the constraints (3) if

1.  $\mathcal{C}_\infty$  is control invariant, and
2.  $\mathcal{C}_\infty$  contains all of the control invariant sets contained in  $\mathcal{X}$ .

**Note:** The geometric conditions for invariance also hold for control invariant sets, i.e.,

$$\mathcal{C} \text{ is control invariant} \Leftrightarrow \mathcal{C} \subseteq \text{Pre}(\mathcal{C}) \Leftrightarrow \text{Pre}(\mathcal{C}) \cap \mathcal{C} = \mathcal{C}$$

46

**Algorithm 2:** Computation of  $\mathcal{C}_\infty$ **Input:**  $f, \mathcal{X}, \mathcal{U}$ **Output:**  $\mathcal{C}_\infty$ 

1. Set  $k = 0$ .
2. Let  $\Omega_k = \mathcal{X}$ ,
3. Let  $\Omega_{k+1} = \text{Pre}(\Omega_k) \cap \Omega_k$
4. If  $\Omega_{k+1} = \Omega_k$ , then  $\mathcal{C}_\infty \leftarrow \Omega_{k+1}$
5. Else, increment  $k = k + 1$ , go to Step 3.

This algorithm generates a set sequence  $\{\Omega_k\}$  satisfying  $\Omega_{k+1} \subseteq \Omega_k$  for all  $k \in \mathbb{N}$ , and terminates when  $\Omega_{k+1} = \Omega_k$  so that  $\Omega_k$  is the maximal control invariant set for  $x(t+1) = f(x(t), u(t))$  subject to constraints.

**Notes:**

- If  $\Omega_k = \emptyset$  for some  $k$ , then  $\mathcal{C}_\infty = \emptyset$ .
- In general, the algorithm may never terminate. If this is the case, in general  $\mathcal{C}_\infty \neq \lim_{k \rightarrow \infty} \Omega_k$ .
  - Sufficient conditions for  $\mathcal{C}_\infty = \lim_{k \rightarrow \infty} \Omega_k$ :  
 $f(x, u)$  is continuous, and the polyhedral constraint sets  $\mathcal{X}$  and  $\mathcal{U}$  are bounded.
- If the algorithm does terminate, then the set  $\mathcal{C}_\infty$  is said to be *finitely determined*. 

**MATLAB:**

```
model = LTISystem('A', 'B', B);
model.u.min = -5;
model.u.max = 5;
model.x.min = [-10; -10];
model.x.max = [ 10; 10];
Cinf=model.invariantSet();
```

47

**Example: Control Invariant Sets**

Consider the following second order unstable system

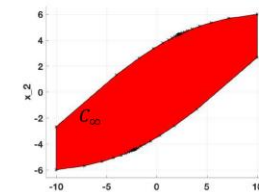
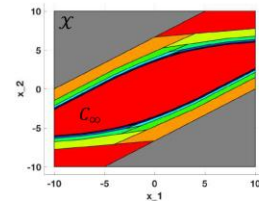
$$x(t+1) = f(x(t), u(t)) = \begin{bmatrix} 1.25 & 0 \\ 1 & -1.5 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t),$$

Subject to state and input constraints

$$x(t) \in \mathcal{X} = \left\{ \begin{bmatrix} -10 \\ -10 \end{bmatrix} \leq x \leq \begin{bmatrix} 10 \\ 10 \end{bmatrix} \right\},$$

$$u(t) \in \mathcal{U} = \{-5 \leq u \leq 5\}$$

**Note:** Algorithm 2 terminates after 36 iterations. In this example, the polyhedron  $\mathcal{C}_\infty$  has 68 vertices.



Top: Iterations of Algorithm 2 to obtain the maximal control invariant set  
Bottom: The maximal control invariant set

48

## The Maximal Controllable Set

For all states in the maximal control invariant set  $\mathcal{C}_\infty$ , there exists a control law such that system constraints are never violated. However, there is no guarantee that the states will be driven to a desired target set. To address this deficit, we need to define two more sets.

**Definition:** For a given target set  $\mathcal{S} \subseteq \mathcal{X}$ , the **maximal controllable set**  $\mathcal{K}_\infty(\mathcal{S})$  for the system (2) subject to constraints (3) is the union of all  $N$ -step controllable sets  $\mathcal{K}_N(\mathcal{S})$  contained in  $\mathcal{X}$ . ( $N \in \mathbb{N}$ )

*For all states in  $\mathcal{K}_\infty(\mathcal{S})$  there exists a control sequence that will drive the system to  $\mathcal{S}$  while satisfying constraints.*

49

## The (Maximal) Stabilizable Set

**Note:** While we know that all states  $\mathcal{K}_\infty(\mathcal{S})$  will reach the target set  $\mathcal{S}$  at some future time, there is no guarantee that once it has reached  $\mathcal{S}$  the system will stay there. *This guarantee can be made if the set  $\mathcal{S}$  is control invariant.*

**Definition:** For a given control invariant set,  $\mathcal{S} \subseteq \mathcal{X}$ , the  **$N$ -step (maximal) stabilizable set** of the system (2) subject to constraints (3) is the  $N$ -step (maximal) controllable set  $\mathcal{K}_N(\mathcal{S})$  ( $\mathcal{K}_\infty(\mathcal{S})$ ).

*For all states in the maximal stabilizable set  $\mathcal{K}_\infty(\mathcal{S})$ , there exists a control sequence that will drive the system to  $\mathcal{S}$  and then cause the system to remain in  $\mathcal{S}$  for all future time-instants while satisfying system constraints.*

50

**Algorithm 3:** Computation of  $\mathcal{K}_\infty(\mathcal{S})$ **Input:**  $f, \mathcal{X}, \mathcal{U}, \mathcal{S}$ , ( $\mathcal{S}$  is control invariant)**Output:**  $\mathcal{K}_\infty(\mathcal{S})$ 

1. Set  $k = 0$ .
2. Let  $\mathcal{K}_k = \mathcal{S}$ .
3. Let  $\mathcal{K}_{k+1} = \text{Pre}(\mathcal{K}_k) \cap \mathcal{X}$
4. If  $\mathcal{K}_{k+1} = \mathcal{K}_k$ , then  $\mathcal{K}_\infty(\mathcal{S}) = \mathcal{K}_{k+1}$
5. Else, increment  $k = k + 1$ , go to Step 3.

This algorithm generates a set sequence  $\{\mathcal{K}_k\}$  satisfying  $\mathcal{K}_{k+1} \subseteq \mathcal{K}_k$  for all  $k \in \mathbb{N}$ , and terminates when  $\mathcal{K}_{k+1} = \mathcal{K}_k$  so that  $\mathcal{K}_k$  is the maximal stabilizable set for the constrained system  $x(t+1) = f(x(t), u(t))$  and the control invariant target set  $\mathcal{S}$ .

**Notes:**

- Because  $\mathcal{S}$  is control invariant, it follows that  $\mathcal{K}_k(\mathcal{S})$  is control invariant for all  $k$ .

51

## Material Covered

- In this module, we have covered:
  - Ch. 4.1-4.4 & 10.1-10.2 of book "Predictive Control for Linear and Hybrid Systems."
- Other references:
  - Kolmanovsky, Ilya, and Elmer G. Gilbert. "Theory and computation of disturbance invariant sets for discrete-time linear systems." Mathematical problems in engineering 4.4 (1998): 317-367.
  - Raković, S. V., Fernando ACC Fontes, and Ilya V. Kolmanovsky. "Reachability and Invariance for Linear Sampled-data Systems." IFAC-PapersOnLine 50.1 (2017): 3057-3062.
  - Blanchini, Franco. "Set invariance in control." Automatica 35.11 (1999): 1747-1767.
- Slide Acknowledgment:
  - Ilya Kolmanovsky @ U Michigan
  - Francesco Borrelli @ UCB

52